toycam

If you've heard that the prices of digital cameras are falling, you've heard correctly. Not only are they cheaper than ever before, manufacturers are also bringing out models that are so basic that they can hardly be called low-end. Instead, we chose to call them toy cam(era)s. Toy cams are basic, no-frills performers that strive to provide decent picture quality and features at the lowest possible price—most of them can be used as Web cameras as well. These cameras are meant for users who just want to take pictures and transfer them on to their PCs, and cost less than Rs 15,000.

Another breed of cameras are the pseudo full-size models, which look like full-sized digital cameras, but don't offer the advanced features mandatory to that kind. These are the gadgets that most folks have been heading towards, mainly for the consideration they show for the buyer's pocket.

So are these toy cams just playthings, as their name seems to suggest? Is their only allure their price? Exactly what we intend to find out!

The Olympus C150 is ergonomically well designed

Test Analysis

We tested these cameras on various parameters such as the features offered, size, weight, overall ergonomics of the camera and performance. *(See box, 'How We Tested').* For performance, we took into account the details captured in a photograph, the colour reproduction, and the brightness and contrast levels of the final output.

Features

Let's zoom into the various features that these cameras offered. The picture resolution is of vital importance to get a crystal clear image—the higher the resolution supported, the better the quality of photos. In this regard, the twin cameras from BenQ—the DC3410 and the DC2300, the PC-CAM 850 from Creative and the Olympus C150 support a resolution of 2,048 X 1,536—excellent for toycams.

Cameras with optical zoom are better for distant photography, say when trying to capture a distant house on the edge of a cliff. But none of these cameras come with optical zoom—instead they try to compensate for lost ground with digital zoom. Unfortunately, using digital zoom compromises picture quality, so one can only use it only when it is absolutely necessary. The BenQ siblings and the PC-CAM 850 were the only cameras with 4X digital zoom. All the others either had no zooming capabilities, or were limited to 2X.

Auto focus and macro focusing are other important parameters that you should keep in mind when buying a camera. Auto focus helps you focus on a subject automatically, but for a camera lacking it, you need to move either forward, or backward, to focus. The same applies to macro focusing, where your subject is extremely close to your lens, and you need to take a perfect picture without any blurring. The Olympus C150 has auto focus as well as a macro focus range of just 20 to 50 cms, thus helping you to take the perfect picture. The creative PC-CAM 850 was the only camera without focus features, making it a typical point and shoot camera. Only what's in the focus will appear in the image, resulting in a very flat image with hardly any depth of field.

All the cameras had white balance settings, except for the Frontech Wondercam and Logitech Click smart 510. All came with four preset white balance options, and you can choose one depending on the environment you're shooting in. They are better for shooting indoors, where one has the option to switch from fluorescent to tungsten, depending on the lighting in the room.

All the cameras store images in the compressed JPEG format, thus saving on space while preserving image quality.

Physical features and ergonomics

Physical features play an important role in the ergonomics of any handheld device, even the camera—the placement of the flash, lenses and light sensors must be such that they are not obstructed by your fingers. Unlike other portable gadgets, cameras can be too light. The perfect camera has to have just the right balance of size and weight—heavy enough to keep your hand steady when shooting, and small enough to be pocketable.

The Olympus C150 Camedia was the best camera in this segment, as far as physical features and ergonomics are concerned. It fits perfectly in your palm, and the contoured front lens cover provides a good grip and stability, making up for the lack of a rubber grip. Another camera worth mentioning here is the PC-CAM 850. Its small size makes it ideal for one-handed operation, which makes it great companion for outdoor shooting. The cameras from BenQ were slim, long and pocketable, while the cam-

What is depth of field?

Depth of Field (DOF) determines the capturing of multiple subjects with the same lucidity, as the prime subject on which the focus is set. You can achieve a better DOF by tweaking the aperture, f-stop and zoom. These settings are not provided in a normal point-and-shoot camera. The lack of these setings make the pictures look flat.

Smaller the aperture (opening of the overlapping blade) size, greater will be the DOF. f-stop is an adjustment that controls the lens' diameter. The f-stop number increases with the decrease in aperture size. Many cameras allow you to manually set the f-stop to enhance the DOF.

A small aperture size allows less light to hit the CCD. Therefore, set a slower shutter speed, preferably to the factory settings for sufficient light to get in through the lens so that the image is not underexposed.

Since the aperture opening will be less, the amount of light hitting the CCD will also be less. The shutter speed will have to be kept slower (at the factory setting), so that enough light gets in through the lens and image is not left underexposed so as to appear dull.

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